

Final project

Introduction

- NLP helps to solve many problems related to human-computer communication
- During the course we have reviewed some of the most relevant techniques and tasks in NLP
- In recent weeks we have discussed the topic of neural language models such as LLMs and explained their potential use today
- The aim of the final project is to integrate the techniques seen in class, including LLMs, to implement a solution to a specific task related to NLP technologies

Specifications

- Form a team of 3 to 4 people
- The team must propose a solution to the following task:
 - Information retrieval
 - Text classification
 - Sentiment analysis

Requirements

- For the solutions to be valid they must:
 - Use some of the NLP techniques reviewed in class
 - It must use LLMs
 - It is necessary to use metrics to evaluate the performance of the methods
 - The use of Spanish corpus is mandatory
 - A user interface to interact with the implemented solution is also recommended (but not mandatory)

Methodology

- The following methodology is mandatory for the implementation of the project:
 - 1) Project definition
 - 2) Corpus selection
 - 3) Selection of frameworks, libraries or technologies used
 - 4) Specification of normalization steps
 - 5) Specification of text representation
 - 6) Selection of machine learning/deep learning/LLMs methods
 - 7) Definition of evaluation metrics
 - 8) Definition of a baseline to compare model performance

Evidence

- The following evidence must be uploaded to TEAMS platform:
 - Source code
 - Resources like corpus, lexicons, datasets, etc.
 - Report of the project development

Report specifications

- The report must contain the following sections:
 - Introduction. Problem description, relevance of the task, possible applications and proposed solution
 - Corpus and datasets. Corpus and datasets used for the task, specifying:
 - Size, content description, tagged classes (when used)
 - Additional resources. Description of all the resources used as lexicons, pretrained LLMs, etc.
 - Solution. Description of the solution specifying:
 - Text preprocessing
 - Training phase
 - Test phase
 - Experiments and results
 - Description of the experiments performed
 - Evaluation of the training phase
 - Evaluation of test phase
 - Conclusions and future work